

THEIA

The team field guide

Understanding the science, evidence and next steps

For the students preparing for ISEF and JSHS | 7 October 2026

The question at the center of the project

When we teach an AI model to put its attention on a lung lesion, does that make its prediction of a cancer-associated mutation more trustworthy?

This guide explains what THEIA actually tests, how its experiments work, what its historical results show, and why the ongoing corrections matter. It is written for teammates who need to understand the whole project, including the parts they did not personally implement.

01

Locate

Does it point correctly?

A controlled audit of medical imaging AI

02

Intervene

What moves the output?

03

Validate

Does it predict reliably?

Evidence snapshot: current v0.3 manuscript and the completed October 7 integration review. A newer Claude execution phase is running separately. Its unfinished work is not counted here.

Learning document, not submission text. Prepared with OpenAI Codex from the project's records and official competition guidance. All diagrams are educational schematics, not patient images. Study, check the evidence, then explain the project in your own words. This guide does not certify competition eligibility or scientific completion.

How to use this guide

Three reading routes

First team meeting: read pages 3-8, 17-19 and 27. You should leave able to state the question, identify the patients, distinguish the outcomes and name the major limitation.

Methods discussion: add pages 9-16 and 20-26. These explain the model, controls, statistics, corrections and remaining scientific work.

Before judging: work through pages 28-33 together. Close the guide before answering the practice questions. Return to the actual project files whenever an answer depends on a number.

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How statements are labeled

HISTORICAL M0 means evidence from the original inputs and models. Known preprocessing defects limit these results. They remain part of the scientific record, but are not corrected-input results.

PLANNED M1 means the corrected-input replication. In the last completed review, 0 of its 50 fits had been accepted. That is not the same as saying no THEIA experiments have been run.

ILLUSTRATIVE means a teaching example with invented numbers or shapes. Such examples are never research observations.

[P1], [P2] ... identify project sources; **[W1]** ... identify public sources. The final pages contain the exact paths and links. Numbers are transcribed from those records unless explicitly calculated here; this guide does not independently rerun the experiments.

01 | What THEIA is trying to learn

THEIA is currently a methods study about the interpretation of medical imaging AI. Its experimental model receives a small stack of CT images around an already annotated lung lesion. It estimates EGFR mutation status and produces spatial attention maps. The team asks whether teaching those maps to overlap the lesion changes the prediction in a meaningful way. [P1]

The motivating problem

A convincing heatmap can make a model appear trustworthy. If the colored region covers the tumor, a viewer may infer that the model used tumor biology to make the right molecular prediction. That inference bundles several claims together. THEIA tests them separately.

1. **Localization:** does the map overlap the annotated lesion?
2. **Model dependence:** would changing a region or internal representation change the output?
3. **Predictive validity:** does the output distinguish the reference labels and express reliable probabilities?
4. **Clinical utility:** would using the model improve real decisions or patient outcomes?

The current evidence is strongest for the first question. The interventions address bounded versions of the second. Predictive evidence remains limited, and clinical utility has not been established.

What could make this useful science?

A carefully controlled result can show researchers which tests a heatmap passes and which conclusions do not follow. That can improve how future imaging models are evaluated even if THEIA is not the best classifier. The contribution is the combination of matched training, geometric controls, interventions, uncertainty checks, external contours and explicit correction of defects. The paper does not claim to have invented the general idea that attention and prediction can diverge. [P1, P2]

Teach-back: explain why a model that points at a tumor is not automatically a model that predicts a mutation accurately.

02 | The medical and imaging basics

Lung cancer includes different diseases. NSCLC means non-small cell lung cancer; adenocarcinoma is one subtype. The project uses existing research collections, not a screening sample of everyone who might develop lung cancer.

EGFR means epidermal growth factor receptor. The gene encodes a protein involved in cell signaling. Certain alterations in a tumor's EGFR gene can influence treatment selection, including eligibility for particular targeted therapies. Clinical decisions depend on the actual variant, disease setting and validated testing. An imaging score is not a substitute for that information. [W1]

Mutation status is a label supplied by a molecular testing process. THEIA attempts to predict a recorded label; it does not read DNA directly. The relevant changes are in tumor biology, and a positive tumor result is not automatically a statement about an inherited mutation.

What CT gives the model

Computed tomography reconstructs a three-dimensional image from X-ray measurements. A slice is one plane through the volume. A voxel is a three-dimensional image element; a pixel is its two-dimensional counterpart. Values in Hounsfield units describe X-ray attenuation. Windowing maps a chosen intensity range to visible brightness.

A **lesion** is an abnormal region. A **segmentation** or contour outlines its extent. A **point annotation** gives a location without the full boundary. Neither is itself an EGFR label.

Radiogenomics studies relationships between imaging features and genomic characteristics. An association can arise through tumor biology, patient composition, acquisition differences or confounding. Establishing which explanation is plausible requires experiments and appropriate comparisons.

What this project does not yet offer

It is not a validated whole-CT cancer detector, screening system, treatment recommender, biopsy replacement or clinical diagnostic service. A lesion annotation is already needed to construct its inputs. [P1]

Teach-back: explain the difference between a CT image, a tumor mask and an EGFR reference label.

03 | Four claims that must stay separate

LOCALIZATION

Map overlaps
the lesion

DEPENDENCE

Changing information
changes the output

PREDICTION

Scores agree
with reference labels

UTILITY

Using the score
helps a real decision

Each box needs its own experiment. None automatically proves the next.

Localization is a geometric claim

A high pointing score says that the map's highest-valued location is often inside the target. It does not say the output is accurate or that only that region supplied information. Centered crops make some localization tasks easy before the model sees image content.

Dependence is an intervention claim

If an output changes after a region is edited, the model responds to that intervention. The edit may remove useful information, add an artifact or disrupt context. A good control helps isolate those alternatives, but no single edit proves exclusive dependence on tumor tissue.

Prediction is a label-based claim

AUC measures ranking against reference labels. Log loss and Brier score assess probabilities. A model may rank people reasonably while producing probabilities that are much too extreme. A visually good explanation cannot repair that.

Utility is a decision-based claim

Useful care would require evidence about a specified decision, appropriate comparators, error costs, calibration, transportability and ultimately the intended clinical setting. THEIA has not tested that chain.

A useful mental check: a highlighter can accurately mark the relevant paragraph even when the answer written underneath is wrong. Conversely, an answer can be right for a reason the highlighter does not reveal. This is an analogy, not an experimental finding.

Sources: [P1, P2].

04 | The data: count people correctly

Collection	Role in the project	Key denominator
NSCLC-Radiogenomics	Internal development and historical controlled experiments	158 built inputs; 153 known EGFR labels, including 40 mutant
RIDER Lung CT	Independent contours and repeat-scan behavior	28 people, two scans each; no EGFR endpoint
TCGA-LUAD v1	Small external molecular evaluation	40 evaluated people: 4 positive, 36 research-sequencing negative
CPTAC-LUAD and unused TCGA cases	Future external work requiring qualification	Candidates, not a completed validation cohort
NSCLC-Radiomics	External localization in the older study	420 patients reported historically; not 420 new EGFR labels

The internal patient flow

NSCLC-Radiogenomics contains 211 source subjects. The current local inventory has 172 with a two-class clinical EGFR status. The built input set contains 158 patients: 153 known labels and 5 unknowns. Unknown does not mean negative. Of the built set, 119 have lesion extent annotations and 39 have point annotations. These are overlapping descriptions of the same people, not additive cohorts. [P1, P6]

The 40 mutant patients represent about 26% of the 153 labeled patients. A study with many image slices but only 40 positive people still has only 40 positive independent patient units for this endpoint.

The unit depends on the question

An image edit generates multiple observations for the same patient. Five folds generate different fitted models. A repeated scan gives another measurement of one person. A bootstrap sample is a computational resample. None of these creates a new participant.

For RIDER, preserve each person's test-retest pair during resampling. For internal prediction, use patient-level out-of-fold predictions. Do not split one person's slices across training and testing.

Sources: [P1, P2, P6, P8]. Counts describe the cited snapshots, not a new data census.

05 | Labels, confounding and leakage

The label is a measurement, not perfect truth

The internal EGFR reference is a clinical-record result from a targeted SNaPshot assay at tested positions in exons 18-21. A recorded wild-type result means no mutation was detected at those tested positions. The released label does not establish that every possible variant was excluded. [P1]

The external TCGA endpoint concerns qualifying activating kinase-domain variants in open research sequencing. A research-sequencing negative is not automatically a clinically verified wild-type result: assay coverage, filtering and variant definitions differ. The project has not demonstrated full equivalence between the internal and external endpoints.

Site can act as a shortcut

In the labeled internal set, one institution contributed 32 mutant patients out of 81, while the other contributed 8 out of 72. Institution also covaries with smoking, sex, scanner vendor, sub-cohort and annotation type. A model can exploit a correlated signal without learning the biological relationship the investigators hoped for. [P1]

The AMC prefix denotes a Stanford sub-cohort, not a third hospital. Annotation tier, sub-cohort and site must not be treated as interchangeable independent factors.

Confounding is not the same as leakage

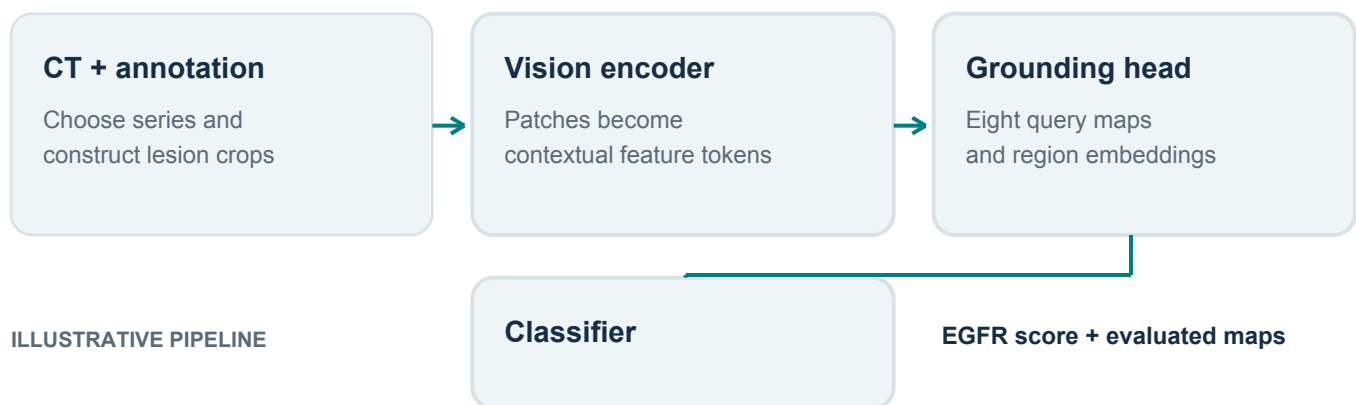
Confounding: a factor such as institution is related to both the inputs and the label, complicating interpretation.

Leakage: information that should be unavailable during training, tuning or preprocessing influences the fitted system or evaluation. Examples include test-set normalization, selecting the best seed after inspecting test outcomes, or the same patient appearing in both partitions.

A site-only baseline is a useful warning: historical imaging AUCs did not exceed it under the same rank-normalized estimator. This does not prove which image features the network used. [P2]

Teach-back: why is excluding unknown labels different from calling them wild type?

06 | From CT to a prediction



A deliberately restricted input

The current manuscript describes 16-slice inputs, each resized to 224 x 224 pixels. Crops follow the annotated lesion and include surrounding context. Per-slice crop boxes and offsets matter: the model sees the selected crop, not the entire original CT volume. [P1]

A feature extractor plus a learned readout

BiomedCLIP ViT-B/16 is the pretrained vision encoder. Its last two transformer blocks are fine-tuned in this study. A transformer turns image patches into feature vectors, or tokens, and allows information to mix across locations. The grounding head uses eight learned queries to attend over a 14 x 14 spatial grid after slice pooling. A classifier reads the resulting region embeddings and outputs a score. [P1]

Because encoder tokens have already exchanged information, a token at a tumor position can carry information about surrounding tissue. A spatial location is therefore not a guarantee of exclusively local content.

What belongs to the current paper

The experimental classifier contains EGFR and KRAS tasks, but EGFR is the reported target here. Older project documents describe a language-generation/rationale branch and broader multimodal ambitions. Those descriptions are not evidence that a clinical report generator or pathology system has been validated. The current scientific argument concerns spatial grounding and EGFR prediction. [P1, P8]

Teach-back: draw this pipeline without notes and mark where the annotation enters.

07 | How attention and grounding work

A query is a learned vector used to select information from patch tokens. Conceptually, the head compares a query with patch keys, turns the match scores into attention weights, and forms a weighted sum of patch values. A region embedding is the resulting summary. The classifier converts one or more summaries into a prediction. [P1]

The model's two main training arms

G0: classification training with grounding-loss weight zero.

G50: classification training plus a grounding loss with weight 0.5. The spatial target is a projected lesion mask. In the original arms, supervision is applied to the pointwise maximum over eight query maps, while the classifier reads the mean of the eight region embeddings.

The 0.5 is a loss weight. It is not a claim of 50% attention, 50% accuracy or a 50% probability threshold.

Why the mismatch is scientifically interesting

The union map can overlap the lesion if one query focuses there. The classifier may also receive information from the other seven queries. Therefore, supervising the union and reading the mean do not enforce a single identical information pathway.

Different reported maps answer different questions:

Map or statistic	What is combined
Query-0 map	One designated learned query
Mean map	Average of query maps
Union map	Pointwise maximum across query maps
Any-query pointing	Best pointing score across the individual queries

A maximum over eight maps has more opportunities to hit a target than one map. Comparators and chance references must match the statistic. Grounding a map does not prove that ungrounded queries are irrelevant, or that a grounded query uses only lesion content.

Source: [P1, P2].

08 | Training, validation and fair comparisons

Five-fold patient-level cross-validation

Divide patients into five outer groups. Train on four groups and predict the fifth. Repeat so each person receives an out-of-fold prediction from a model that did not train on that person. The pooled predictions support one patient-level evaluation. A separate inner validation split selects checkpoints without using the outer test fold. [P1]

The original G0/G50 arms selected epochs using a composite of EGFR AUC and grounding mass lift. This selection rule rewards localization in both arms. The matched N08 experiment instead selected the minimum inner-validation EGFR log loss, with earlier epochs winning ties. Comparisons must acknowledge this design difference.

Seeds are repeated fits, not new cohorts

Historical configurations 1337, 42 and 7 changed partition and initialization together. Their results reveal sensitivity to those combined choices. They reuse the same patients and cannot be pooled as three independent clinical studies.

The planned corrected replication separates these choices more clearly: its partition seed is fixed, with two initialization seeds. Two seeds still do not establish the full distribution of training randomness. [P5]

What makes a comparison matched?

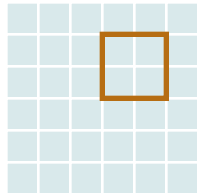
Keep the patients, labels, folds, preprocessing, selection rule and relevant training settings aligned. Then change the factor under investigation. Otherwise, an apparent effect of grounding may actually reflect a different split, checkpoint rule or patient set.

A model that fails numerically is a failed fit. Do not silently discard it, choose a more favorable retry or average away nonfinite values. Preserve the failure and follow the predefined retry and acceptance rules.

Teach-back: why must checkpoint selection happen inside the outer training boundary?

09 | Measuring where the model attends

Uniform reference



Center prior

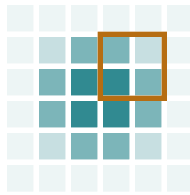
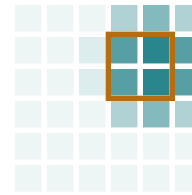


Image-dependent map



Gold outline = toy target. These grids are schematics, not THEIA observations.

Pointing asks whether a highest-valued map location lies within the projected target. Production uses a tie-aware rule: it averages the hit over cells within the specified tolerance of the maximum. If four maximum cells tie and only one is in the target, the score is 0.25, not automatically 1. [P1]

Attention mass is the fraction of normalized map weight inside the target. A uniform map assigns mass equal to the target's area fraction. If a target covers 4% of the grid, roughly 4% mass is its uniform reference, not evidence of localization skill.

Area-matched IoU selects as many map cells as the target contains, then measures intersection divided by union. If selection-boundary ties occur, multiple equally valid selections can produce different overlaps. Historical production ordering is one choice; admissible bounds describe the ambiguity. [P7]

Why the baseline matters

A fixed center-peaked map sees no image content but can perform well on lesion-centered crops. A spatial shuffle tests a different reference. An empirical prior must be constructed from training-fold masks only. Always name the comparator and use the same scoring rule for both maps.

A score is also conditional on the grid and projection. Reducing a 3D lesion to a slice-pooled 2D target loses location information. Scores across different grids or target constructions are not automatically comparable.

Teach-back: why can a uniform map and a center-peaked map have very different chance scores?

10 | Measuring prediction and uncertainty

AUC is a ranking measure: the probability that a randomly selected positive receives a higher score than a randomly selected negative, with half credit for ties. An AUC of 0.66 is not 66% accuracy and does not specify a usable decision threshold.

Log loss penalizes incorrect probabilities, especially confident mistakes. **Brier score** is the mean squared difference between a probability and a binary label. Lower is better for both. **Calibration** asks whether stated probabilities match observed frequencies; good ranking alone does not ensure it.

Which AUC is being reported?

Raw pooled out-of-fold AUC compares scores from multiple fold models on their original scale. Within-fold rank-normalized pooling changes those scores before pooling. These estimate different quantities. N08's primary contrasts use raw pooled AUC; the historical site-only comparison uses rank-normalized pooling for both models. Do not substitute one table for the other. [P2-P5]

Read an interval, not just its sign

A historical N08 contrast is +0.077 AUC with a 95% interval from -0.014 to +0.168. This estimate is compatible with a small adverse change and a meaningful benefit under its stated uncertainty model. It does not establish improvement, equivalence or no effect.

These intervals largely resample patients while holding fitted models fixed. They do not automatically include retraining, new institutions or every analytic choice. A frequentist 95% procedure aims to cover its target in repeated use; a single realized interval is not a 95% probability statement about a fixed true value.

Multiple questions create multiple chances

Testing many contrasts makes an isolated low p-value less persuasive. Label exploratory analyses, define the primary question before new outcomes and use a validated multiplicity procedure when claiming family-level error control. More bootstrap draws improve computational precision; they do not add patients.

11 | The experiment map

Experiment family	Main question	What it cannot establish alone
Geometry controls	Does localization exceed a plausible image-free map?	EGFR accuracy or biological dependence
Matched training: N08	What changes when grounding/readout changes?	Broad generalization from one cohort
Input edits: N02/C22	Which controlled image perturbations change output?	Tumor-exclusive information or clinical benefit
Component randomization: N03	What happens when one fitted component is replaced?	Necessity after retraining
Attribution deletion / IG	Do ranked regions identify output-sensitive edits?	A unique biologically correct explanation
Value-donor interventions: E33/N04	Does the output follow substituted internal content?	Input-space localization of the causal signal
RIDER repeat scans	Are outputs/localization stable under reacquisition?	Molecular correctness without labels
External TCGA test	How does a fixed model perform in a new collection?	Precise transfer estimates with four positives

Why use more than one experiment?

Each experiment breaks a different possible shortcut in the argument. A geometric baseline challenges crop bias. Matched training challenges confounded model comparisons. Input edits challenge a purely visual explanation. Randomization challenges attribution to a particular learned module. External data challenge transportability.

Agreement among several tests can strengthen a narrow conclusion. It cannot erase shared limitations: many analyses still reuse the same patients, same trained models and historical inputs.

Experiment codes are bookkeeping identifiers. A long list of completed codes is not itself a measure of scientific quality. The important question is which uncertainty each experiment reduces and whether its evidence changes a claim.

Sources: [P1, P2, P7].

12 | Input edits and their controls

Imagine replacing part of an image with blur or noise and observing the change in the EGFR logit. The experiment compares the lesion edit with matched-size control edits in the same patient. Pairing helps remove baseline differences between patients. [P1]

A control is a scientific choice

Scattered controls distribute edited cells or voxels across locations. **Translated controls** move the lesion's own shape elsewhere. Other constructions avoid the lesion or use distant locations. Matching the number of edited elements does not necessarily match tissue type, texture, contiguity or artifact severity.

The target itself can be defined by a projected patch-grid mask or the native-resolution lesion mask. Those interventions modify different inputs. Their signs need not agree.

What the historical experiments found

Under noise fill, projected-target and native-mask edits had opposite-signed contrasts against scattered controls in the supervised arm. That pattern recurred across the additional training configurations under the frozen reading rule. The lesion-versus-translated-control effect also recurred. Blur results were less consistent. [P1, P2]

This means the apparent lesion effect depends on the precise intervention and comparator. It does not mean that one sign proves the tumor protects against mutation or that the other proves it causes mutation. The experiment changes model inputs, not patients' biology.

Output movement is not necessarily worse prediction

A large change in a logit shows sensitivity. To ask whether that change harms prediction, compare against the reference labels using a proper loss. In the historical label-based edit contrasts, only one of 24 log-loss intervals and two of 24 Brier intervals excluded zero; these were uncorrected comparisons. [P2]

Teach-back: why is an edited image potentially off the distribution on which the model was trained?

13 | Inside the model: components and values

Replacing one component at a time

N03 re-initializes a selected component while freezing the rest of a fitted network. Tested components include query vectors, query/key/value projections, the output projection, the entire grounding head and parts of the encoder. Replacements are non-cumulative: each comparison starts from the original model. [P1]

A replacement draw is shared across patients. Consequently, errors introduced by that draw are correlated. Treating every patient's response as independent would understate some uncertainty. The revised analysis separates patient variation and shared-draw variation.

Historical query-0 pointing fell from about 0.94-0.97 to 0.42-0.46 after whole-head replacement. Yet some localization remained, and supervised encoder tokens behaved differently from unsupervised tokens under the randomized readout. This supports a bounded statement about where learned representation and readout effects reside. It is not a complete biological decomposition. [P2]

Swapping the value pathway

The value-donor experiment replaces a recipient's internal value tensor with a compatible donor's tensor while preserving the specified remainder of the forward path. The question is whether the output follows the donated content. Controls must first show identity preservation, a connected nonzero positive response and the intended negative response on the actual model path.

The reported historical mean absolute probability movement was about 0.30. A donor-following statistic was about 0.93 under its population-oriented estimator, with limitations in interval validation. It is not a percentage of tumor causality or a measure universally bounded by one. [P7]

An unresolved precision issue

In N03, the prespecified Monte Carlo precision criterion was unmet for 106 of 300 primary estimates. Corrected intervals include shared-draw uncertainty, but a failed precision target must remain visible. Additional random draws would address simulation noise, not small patient numbers or model retraining uncertainty.

14 | Attribution, deletion and query traps

Saliency uses local sensitivity of the output to input changes. **Integrated gradients (IG)** accumulates gradients along a path from a chosen reference input to the actual input. Both depend on the model, target output and implementation. An attention map is another object; it is not interchangeable with either. [P1]

Test the ranking with deletion

Rank cells by an attribution map, edit increasing numbers of cells, and record output movement. Compare the resulting curve against random or geometric rankings using the same edit operator. A curve depends on both the ranking and the fill, so blur and noise need not give the same answer.

Historically, IG exceeded random under blur deletion, but not under noise. Attention's blur comparison with random had an interval that included zero. The direct IG-minus-attention comparison also included zero. Therefore, "IG was significant and attention was not" does not establish that IG outperformed attention. [P2]

Numerical convergence is necessary, not sufficient

More integration steps changed many patient-level deletion selections. The cohort-level contrasts moved little, but one supervised patient remained unresolved at the maximum checked steps. A small sum-level completeness error can also conceal cancellation between component errors. Report what was checked and what remains uncertain.

Beware an attractive query story

A concentrated query can look special. But comparing one query with a group of seven mixes contribution with group size. Leave-one-out and Shapley-style shares depend on the replacement game, baseline and pooling geometry. Earlier THEIA interpretations were corrected after checking these properties. [P7]

The lesson is general: define the estimand before giving a statistic a biological interpretation. Reproducing the number is only the first check; validate what the number means.

15 | The matched N08 design

N08 was designed to separate the readout choice from spatial supervision. A cell is one training condition. Historically, each of five cells was trained on five folds, giving 25 fits on one partition and initialization. All reported results on this page are **HISTORICAL M0**. [P3]

Cell	Classifier readout	Grounding target	Raw pooled AUC	Readout pointing
A	Mean of queries	None	0.586	0.017
B	Mean of queries	Union, weight 0.5	0.663	0.571
C	Query 0	None	0.586	0.025
D	Query 0	Union, weight 0.5	0.531	0.227
E	Query 0	Query 0, weight 0.5	0.590	0.723

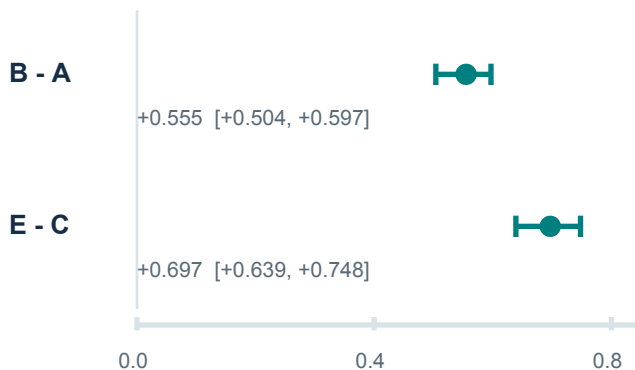
Prediction uses 153 labeled patients, including 40 positives. Spatial scoring uses the historical 119 with projected targets. Rounded cell scores may not subtract to exactly the published paired differences.

Read the comparisons precisely

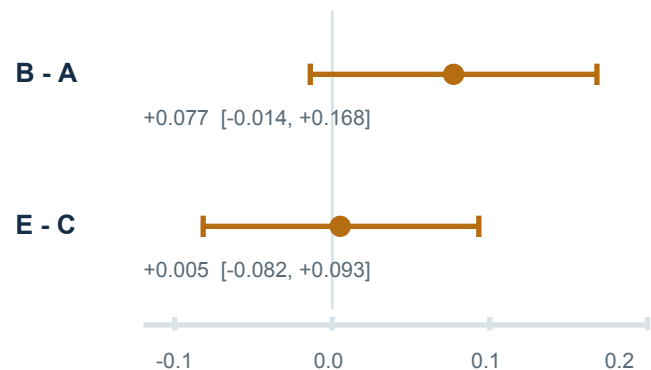
- B minus A:** add union grounding while reading the mean. Pointing improves strongly; estimated AUC change is +0.077.
- E minus C:** add grounding directly to the query used for prediction. Pointing improves strongly; estimated AUC change is +0.005.
- D minus C:** add union grounding while reading query 0. Estimated AUC change is -0.054. This is a different contrast from E minus C.
- E minus D:** change where supervision is applied while keeping query-0 readout.
- Interaction:** (D - C) - (B - A) asks whether the effect of union grounding differs between the two readouts. Its historical AUC estimate is -0.131. It became an exploratory lead after outcomes were seen and did not survive the cited post hoc multiplicity adjustment. It is not an accepted replicated finding. [P2, P3]

16 | The central result, with uncertainty

POINTING DIFFERENCE



AUC DIFFERENCE



HISTORICAL M0 | 119 spatial targets; 153 labeled patients | 95% intervals

The two panels show paired differences from historical N08. Horizontal bars are the reported 95% patient-bootstrap intervals conditional on the fitted models. Prediction and localization have different units and scales. The vertical line means no difference. [P2, P3]

What the pattern supports

Both matched supervision contrasts substantially increased readout-consistent pointing. In those same comparisons, the AUC intervals included zero. The localization result is therefore much clearer than the predictive result for these fitted systems.

What the pattern leaves open

The mean-readout AUC interval extends to +0.168. The query-0-targeted interval extends from -0.082 to +0.093. Those ranges do not justify saying grounding has no predictive effect. An equivalence claim would require a justified margin and sufficiently precise evidence within it.

The analysis also rests on the historical input version and one matched training configuration. Corrected inputs, repeat fits and independent patients could change the estimates or their interpretation.

Practice distinction: “The experiment did not demonstrate benefit” describes the evidence. “There is no benefit” makes a stronger statement that these results do not establish.

17 | External tests and repeat scans

RIDER: localization and repeatability

RIDER provides 28 patients with paired repeat CT scans and independent manual contours. It has no EGFR endpoint. Historical supervised union pointing was 0.871, 0.864 and 0.850 across three training configurations; unsupervised pointing was 0.239, 0.304 and 0.146. These are repeat fits evaluated on the same 28 people, not three independent cohorts. [P1, P2]

RIDER's center comparator was added post hoc. Scoring it with the same tie-aware rule matters: the “any central cell hits” rule is more generous than averaging over tied central cells.

The supervised model's logit ICC was about 0.80-0.88. Nevertheless, four or five of the 28 patients changed by more than 0.10 in predicted probability in each configuration. That threshold is descriptive, not clinically validated. A high ICC can coexist with large errors for individuals because between-person variation influences it.

TCGA-LUAD: a small molecular test

The historical external test evaluated 40 people: 4 positive and 36 research-sequencing negative. Its AUC was 0.417, with a 95% interval of 0.139-0.759 and exact permutation p about 0.62. Mean predicted probability was 0.26 versus observed prevalence 0.10. [P1, P2]

There are only $4 \times 36 = 144$ positive-negative pairs. Without ties, correcting the order of one pair changes AUC by $1/144$; one positive can affect up to 36 pairs, or 0.25 AUC. This arithmetic illustrates why the result is imprecise.

The test neither establishes nor rules out transfer. The label definition differs from the internal reference, and these outcomes are now exposed. Later analyses of these people are no longer untouched external testing.

Teach-back: which external dataset tests molecular prediction, and which tests localization and reacquisition?

18 | What changed during the revamp

The original study emphasized external attention localization and weak incremental EGFR prediction over clinical variables. The newer paper focuses on a controlled audit of what lesion supervision actually changes. The progression added matched training, stronger geometry controls, input interventions, component randomization, attribution checks, repeat scans and a small molecular external test. [P1, P8]

Corrections are part of the scientific record

Several impressive-sounding interpretations weakened after closer inspection. Query group effects partly reflected group size. A donor-following statistic did not have the assumed universal bound. Shared randomization draws required shared uncertainty. Metric disagreement needed mathematical validation rather than a permissive label. These corrections improved the meaning of the evidence. [P7]

The important input defect

Seven point-annotated internal inputs used an annotation frame number in the wrong CT series, producing crops at the wrong anatomical location. Those cases remained in historical training. Dropping them only from evaluation cannot undo their influence on the trained models. [P1, P2]

SEG-side review also identified misplaced imported masks and sampling that included annotation-empty or small-fragment planes. A corrected import may faithfully preserve every supplied annotated voxel while still producing a scientifically awkward sample. Faithful conversion and clinically sensible sampling are different questions. [P4]

Why old results are not silently replaced

M0 is the historical input lineage. M1 and its S1-a candidate contain corrections under explicit policies. Changing inputs creates a different experimental system. Preserve the old results, document the changes, retrain and compare under a frozen interpretation plan.

A successful corrected run would not erase the defect. It would establish what happens under the revised system and show how much the scientific account changes.

19 | Sampling and geometry: a worked audit

In the nine rebuilt SEG cases reviewed in the completed October 7 phase, 16 sampled positions per patient give 144 positions. Only 134 are unique CT planes because some positions repeat. These denominators answer different questions. [P4]

Position-level category	Count
Annotation-empty positions	23
Positions containing only a secondary component	18
Positions containing the main component	103
Total	144

A separate classification asks where the crop box came from: 35 boxes used minor-component planes and 109 used main-component planes. These are not the same categories as “what annotation is present in the sampled plane.” Do not combine them into one partition.

Why a tiny fragment matters

A small distant annotated component can extend the first-to-last slice range. Sampling uniformly across that range can select many planes away from the main lesion. A crop box borrowed from a nearby annotated plane may then show little or none of the intended target. In one audited case, 11 of 16 sampled positions were annotation-empty. [P4]

Two counts that were being conflated

The historical screen found **10** patients with empty or fragment-only sampled slices. The **nine rebuilt cases** are a different set. Calling both “the nine problematic patients” obscures which operation and input version are being described.

These are image-processing and annotation findings, not a new clinical diagnosis of what every connected component represents. Deciding to remove small components or use reader-corrected contours would be another policy change requiring its own rationale and evaluation.

Teach-back: explain how a converter can preserve annotations correctly yet produce unsuitable crops.

20 | The corrected-input replication

The planned N08 replication has **5 cells x 5 folds x 2 initialization seeds = 50 fits**. The last completed review recorded 0 attempted, 0 completed and 0 accepted corrected fits. That is a snapshot; Claude's ongoing phase may later update it. [P4, P5]

What is fixed in the contract?

The cell definitions remain A-E. Patient identities, labels and outer partition stay aligned. The two initialization seeds are 42 and 7; partition seed is 1337. One held-out prediction per patient is required for each cell-seed combination. The spatial eligibility set must be recalculated from corrected inputs rather than copying the historical denominator automatically.

The primary statistic is the average of two seed-specific AUC interactions:

$$I(\text{seed}) = [\text{AUC}(\text{D}) - \text{AUC}(\text{C})] - [\text{AUC}(\text{B}) - \text{AUC}(\text{A})]$$

$$M = [I(42) + I(7)] / 2$$

Compute seed-specific statistics and then combine them as specified. Do not duplicate patients across seeds or silently average their predictions instead.

What would count as recurrence?

The contract's directional recurrence rule requires both seed interactions to be negative and the upper 97.5th-percentile bound for M below zero. The primary lead was chosen after discovery outcomes were known. This is an outcome-informed replication plan frozen before its new outcomes, not a preregistered original discovery. [P5]

Secondary family-level claims require both the primary gate and a validated error-control procedure. The cited contract lists that procedure validation as unevaluated; until resolved, the relevant secondary results remain exploratory.

Because M1 changes inputs and initializations relative to discovery, an M0-M1 difference cannot isolate the effect of preprocessing alone. A clean estimate of a single repair's effect would need a separately matched design.

21 | Can the cohort reach 10,000?

Ten thousand is the team's expansion goal, not an achieved sample size, a verified public-data supply, or a requirement established by this guide. The user has chosen to prioritize that goal even if journal submission slips. [P6, P9]

Count eligible patients, not downloadable scans

A molecular evaluation needs distinct people with usable lung CT, the right disease and timing, a defined EGFR reference, sufficient label quality, appropriate lesion information and lawful access. A large CT collection without EGFR labels may help representation learning or robustness studies. It does not become a 10,000-person CT-EGFR test cohort.

The October 7 inventory identified additional candidates, but no verified public route to 10,000 eligible paired cases. The number 267 seen in earlier handoffs combines 172 internal clinical labels, 59 TCGA research-label candidates and a 36-person CPTAC projection. That is mixed inventory arithmetic, not a verified eligible count or a ceiling on all public data. [P6, P9]

Near-term external possibilities

At that snapshot, 13 never-scored TCGA cases had been issued for reader work, and 28 CPTAC cases were protected candidates before final labeling and reader qualification. Their combined upper bound of 41 is provisional; currently evaluable new cases were zero. The previously exposed TCGA development/evaluation cases must remain distinct.

How to expand without weakening the endpoint

Keep a cohort registry with source, terms, duplicate checks, patient count, label definition, timing, reader status and outcome exposure. Separate auxiliary CT, development CT-EGFR and untouched external CT-EGFR. Do not treat research negatives as verified clinical negatives or merge incompatible assays without a justified harmonization analysis.

Credentials belong to the approved researcher and dataset access remains collection-specific. A friend's PhysioNet approval is not transferable access. Any future institutional collaboration needs appropriate permissions and documented responsibility.

22 | What strong evidence still requires

Immediate scientific priorities

1. Finish input identity, geometry and sampling decisions, preserving the historical lineage.
2. Complete the corrected fits under the frozen contract and retain failures, retries and acceptance receipts.
3. Recompute paired prediction and spatial readouts; inspect disagreements and uncertainty before revising claims.
4. Carry affected historical claims forward only after their required reruns or explicitly retain them as historical.

Questions that require different evidence

Independent molecular validation: qualify external patients with a defensible EGFR reference, adequate positives and negatives, valid lesion localization and a protected evaluation boundary.

Clinical added value: compare with clinically available variables using a valid development boundary, then evaluate on appropriate external data. A higher image-only AUC is not automatically incremental value.

Robustness and generalization: study acquisition differences, site shifts and relevant subgroups with adequate sample sizes. Many tiny subgroup estimates may only produce unstable stories.

Faithfulness: use controls that isolate the intervention of interest, check sensitivity adequate for a null claim, and avoid turning internal tensor experiments into tumor-exclusive input claims.

Clinical utility: define an intended decision and test whether the model helps it. This is a different research stage, not something a heatmap or test suite can establish.

What cannot be promised now

A journal acceptance probability, a competition award, a scientifically defensible completion percentage or a reliable date for 10,000 paired public cases cannot be inferred from the present record. Publication readiness depends on the final supported contribution, not reaching an arbitrary experiment count.

Sources: [P1, P2, P4-P6, P9]. This page summarizes evidence needs; it does not authorize new compute or acquire data.

23 | Software checks versus scientific checks

Reproducibility asks whether a specified procedure produces the recorded result from the identified inputs. **Validity** asks whether that result supports the intended inference. They overlap, but one does not guarantee the other.

Four examples from this project

A hash matches. This establishes byte identity with a recorded file. It does not establish correct anatomy, trustworthy labels or an appropriate estimand.

A unit test passes. This establishes the tested behavior for its fixtures. A test that duplicates the implementation or checks the wrong value can pass without detecting a real defect.

A result is reproducible. It can still be reproducibly biased by a wrong-series crop, a mismatched comparator or an invalid uncertainty procedure.

Two assistants agree. Their agreement is not independent clinical or statistical validation. They may share the same assumptions and source code.

What a convincing evidence chain contains

A claim points to a clearly defined statistic, patient set and input version. Those point to saved predictions or measurements, configurations, code identity and execution logs. Validation probes test plausible failures, including impossible values, empty projected targets, tie behavior and corrupted outputs. Human interpretation then stays within the tested scope. [P4, P7, P9]

An exclusion must earn its reason. An empty returned metric is not proof of an empty target. A named divergence must satisfy the mathematical conditions for that divergence. In THEIA's earlier audit, merely classifying disagreements accepted injected wrong values; mutation tests exposed the weakness.

Teach-back: give one example of a software check that protects the study and one scientific question it cannot answer.

24 | Claims the team can and cannot make

Evidence-conscious statement	Stronger claim not established
Historical matched training substantially improved lesion pointing.	The corrected model has already replicated that improvement.
The predictive intervals include benefit and harm.	Grounding has no effect, or CT contains no EGFR signal.
Supervised maps align with independent RIDER contours.	RIDER validates molecular prediction.
Some controlled edits change the model's output.	The model exclusively relies on tumor biology.
Internal value replacement moves predictions toward donor behavior.	A specific input region causes the patient's mutation.
The small TCGA test is imprecise and endpoint-limited.	The model definitively transfers, or definitively cannot transfer.
Historical imaging scores did not exceed a site-only baseline.	We proved the network uses only hospital identity.
New public-cohort candidates are being qualified.	We have acquired 10,000 eligible CT-EGFR patients.
Tests and hashes improve traceability.	The science is correct because the suite is green.

The language of a negative result

“Not demonstrated” leaves room for uncertainty. “Equivalent” needs an equivalence design. “Robust” needs a named set of perturbations. “Independent” needs a stated independence boundary: people, sites, annotations, code or random draws.

The project can contribute a useful audit without claiming a biological impossibility or a universally superior explanation method. A strong discussion says what would change the conclusion and what evidence is still missing.

Sources: [P1-P7, P9]. Use this table as a reasoning exercise, not a script to submit or memorize.

25 | What is distinctive about the project?

THEIA's strongest current identity is an evidence audit of an attractive inference: that a lesion-centered explanation makes a molecular prediction credible. The historical experiments show why this inference deserves separate testing. [P1, P2]

The pieces that work together

The matched design separates readout and supervision instead of comparing unrelated models.

Geometric controls reveal how much apparent localization could arise from the crop construction or map aggregation.

Different interventions distinguish localization from sensitivity and show how explanations change with the control or fill.

Uncertainty audits reveal when patient-only intervals miss randomness shared across the cohort.

Independent contours and repeat scans ask whether localization and output stability persist under different measurements.

Visible corrections make the project's boundaries inspectable. Input defects, failed precision goals and exposed test outcomes stay in the record.

A potential contribution is not a priority claim

The project does not establish that it is the first to observe a separation between attention alignment and generalization. That broader issue has prior literature. Novelty must be tied to the precise combination of question, molecular CT setting, controlled design and findings, with primary-source comparison before publication.

Nor is a large experiment count automatically impressive. Ten nearly redundant analyses can be less informative than one well-designed external test. Each new experiment should answer an unresolved question, test a failure mode or estimate a quantity with useful precision.

Team exercise: choose three experiments that most strongly support the central claim. For each, identify its comparator, unit of analysis, result and most important limitation. Then identify one result that could overturn your current interpretation.

26 | Preparing for ISEF and JSHS

ISEF: know the current rules

ISEF teams may have at most three members, and every member should understand the project and contribute meaningfully. The current rules restrict the competition period and require the applicable approvals and documentation. They allow acknowledged AI assistance as a resource but prohibit generative AI from writing the research plan, abstract or poster, or generating citations. Confirm the relevant forms and local requirements with your adult sponsor and affiliated fair. [W2]

The official science rubric allocates 10 points to the question, 15 to design, 20 to execution, 20 to creativity/impact and 35 to presentation; the interview accounts for 25 of those 35. Understanding methods, limitations and each student's contribution matters. [W3]

JSHS: individual presentation of team work

JSHS is an individual competition. A team selects one presenter, who must acknowledge the group's contributions; the same person continues if selected nationally. Use the current Statement of Outside Assistance and verify regional deadlines. Its 2026-27 AI policy permits learning and coding support but forbids AI replacing the student's final question, full design, original analysis or substantive submission writing. [W4]

The oral presentation is at most 12 minutes, followed by up to 6 minutes of questions. Practice explaining evidence within that time in your own words. [W5]

A project-specific issue to resolve now

THEIA's records disclose extensive AI participation in coding, analysis and manuscript drafting. Do not describe that as grammar-only assistance or assume reading this guide makes the work eligible. Bring a truthful contribution and AI-use record to the adult sponsor and regional organizers. They can assess what student work qualifies and whether a distinct student-led extension is needed. Rewriting AI prose alone does not establish student ownership.

This guide is for learning. It is not a research plan, competition paper, abstract, poster or presentation script.

27 | A team understanding plan

Meeting 1: the whole project, without code

Each person draws the CT-to-prediction pipeline and labels the four claims: localization, dependence, prediction and utility. Compare drawings. Agree which datasets answer which questions. Every teammate should be able to explain why 28 paired scans mean 28 people, and why 153 labeled patients is the molecular denominator internally.

Meeting 2: one experiment in depth

Assign one family to each person: matched training, input edits, component/attribution checks, or external/repeatability analysis. Each prepares a one-page explanation in their own words containing the question, changed factor, control, outcome, uncertainty, finding and limitation. Rotate explanations so everyone learns the other families.

Meeting 3: inspect the evidence

Open the N08 tables and trace B-A and E-C to their reported intervals. Check that raw AUC and rank-normalized AUC are not mixed. Open the input-repair crosswalk and explain why evaluation-only exclusions cannot repair training. Read the external label definition together.

Meeting 4: challenge the story

One teammate acts as a skeptical judge. Ask what alternative explanation survives, whether a comparison is post hoc, which patients were actually independent and whether the result has been rerun on corrected inputs. Record unanswered questions as research tasks rather than inventing explanations.

Keep an honest contribution record

For each person, record the work they designed, implemented, checked, interpreted and explained; include dates and evidence. Record mentor and AI assistance alongside it. Do not invent contributions or redistribute authorship merely to match a competition format.

Learning the entire project is a shared goal. Performing every part personally is not required, but each presenter must accurately distinguish their own work from collaborators' work.

28 | Questions you should be able to answer

Answer these aloud before turning to the next page. For numerical answers, name the denominator and input version.

1. What is THEIA's central research question, and what medical problem motivates it?
2. What does the model receive? Does it locate a tumor in an unannotated whole CT?
3. What are the four claims that a heatmap alone cannot combine?
4. Why are there 211 source subjects, 158 inputs, 153 EGFR labels and 119 spatial targets?
5. How do G0 and G50 differ? What does the 0.5 represent?
6. Why does union supervision plus mean readout leave an interpretive gap?
7. What do B-A, E-C and the N08 interaction measure?
8. Why is AUC 0.663 not “66.3% accuracy”?
9. Why does an interval including zero not prove no effect?
10. Why is the site-only rule an important comparator?
11. How can changing a lesion region move the output without proving tumor-exclusive dependence?
12. Why can a random head's union map beat a single geometric map?
13. What can RIDER establish, and what can it not establish?
14. Why is the TCGA test with four positives imprecise?
15. What was wrong with the seven point-annotated inputs?
16. Why is corrected-input retraining necessary even if evaluation exclusions barely move AUC?
17. What does “0/50 corrected fits” mean, and what does it not mean?
18. Why do 10,000 CT scans without suitable EGFR labels not fulfill the paired-cohort goal?
19. Which important decisions were made after outcomes were known?
20. What did you personally contribute, and how did you verify AI-assisted work?

Challenge question: design, in your own words, a result that would make you substantially revise the current conclusion. Explain the required data, comparison and uncertainty. Discuss it with the team and mentor before turning it into a new study.

29 | Answer checks, not a presentation script

1-3. The study tests whether lesion-grounding supervision makes attention align with lesions and whether that alignment is evidence about EGFR prediction. Inputs are annotation-centered CT crops. Localization, dependence, prediction and utility need separate evidence.

4. These are successive or overlapping eligibility descriptions. Five built inputs lack a two-class EGFR label; 39 carry a point rather than an extent annotation. Do not add these groups as independent cohorts.

5-6. G50 adds weight-0.5 spatial supervision to the union; G0 removes that loss. The classifier's mean readout still includes other query embeddings. Union alignment does not force all information used by the classifier to be tumor-specific.

7. B-A adds union grounding with mean readout. E-C adds query-0 grounding with query-0 readout. The interaction compares D-C against B-A, not E-C against B-A.

8-10. AUC ranks positives against negatives; accuracy needs a threshold. A broad interval including zero can include meaningful effects. Site predicts labels in this confounded cohort and provides a demanding non-imaging reference.

11-12. An edit also changes texture and distribution. A union combines multiple opportunities to localize; its chance behavior differs from a single map.

13-14. RIDER supports localization and reacquisition analysis, not EGFR validity. Four TCGA positives give only 144 positive-negative pairs and large sensitivity to individual positives, alongside endpoint mismatch.

15-17. Annotation frame numbers were applied in the wrong CT series. Removing evaluation cases cannot undo training influence. Zero accepted corrected fits describes a pending replication, not the absence of historical experiments.

18-20. The expansion needs unique eligible CT-EGFR patients, not image files alone. The interaction lead and several comparator/analysis amendments were outcome-informed. Contributions and verification must be answered from your actual records, not inferred from this guide.

If your answer omits uncertainty, the control or the evidence version, it is not yet complete.

30 | Glossary: the clinical and model terms

Term	Meaning in this project
EGFR	Epidermal growth factor receptor; the gene whose recorded mutation status is the principal target
NSCLC / LUAD	Non-small cell lung cancer / lung adenocarcinoma
Radiogenomics	Study of relationships between imaging and genomic characteristics
Reference standard	The measurement process used to define the comparison label
Wild type	No qualifying mutation detected under the particular assay definition; not a universal absence claim
CT series	A coherent set of CT images from an acquisition/reconstruction; frame numbers are not interchangeable across series
DICOM	A standard for medical images and their metadata
ROI / segmentation	Region of interest / a delineated spatial extent
AIM / SEG	Annotation formats used for point information or segmentation objects in the project's import paths
Voxel / slice	A 3D image element / one image plane
Patch / token	A small image region / its numerical feature representation
Encoder / backbone	The model part that converts images into learned features
Query / key / value	Representations used to choose and aggregate information in attention
Grounding	Training or assessing spatial alignment with an annotated target
Readout	The rule by which internal representations feed the prediction
Logit	An unbounded score associated with probability; in binary form, log odds
G0 / G50	Original arms with grounding-loss weights 0 and 0.5
M0 / M1 / S1-a	Historical inputs / corrected-input lineage / a specific candidate repair and sampling policy

These definitions are learning aids. Exact operational definitions belong to the experiment contract and source code.

31 | Glossary: the evidence terms

Term	Meaning in this project
Estimand	The precisely defined quantity an analysis intends to estimate
Out-of-fold (OOF)	Prediction made for a patient by the model that held that patient out of training
Bootstrap	Resampling the observed units to estimate sampling uncertainty
Paired contrast	Comparison that preserves the correspondence of the same patients across conditions
Calibration	Agreement between stated probabilities and observed event frequencies
Confounding	A correlated factor complicates interpretation of an association
Leakage	Information crosses a boundary that should protect training or evaluation
Ablation	A controlled removal or change to part of a system
Off-distribution	An input or internal state differs from what the model was trained to handle
Monte Carlo error	Uncertainty caused by using a finite number of random simulations or draws
ICC	Intraclass correlation; here an absolute-agreement repeatability statistic on logits
Equivalence	Effects sufficiently close under a justified, predefined margin; not merely a nonsignificant difference
Prospective / post hoc	Specified before the relevant outcome / selected or changed after seeing it
Exposed cohort	A cohort whose relevant outcomes or evaluation results have already been inspected
Transportability	Performance under a different patient, site or acquisition distribution
Provenance	The record linking artifacts to inputs, code, decisions and execution
Claim boundary	The population, system, outcome and uncertainty within which a statement is supported

Final self-check: can each teammate explain a result, identify one surviving alternative explanation and show where the evidence is stored? If so, the guide has served its purpose.

32 | Where to find the evidence

Paths below are relative to **/Volumes/THEIA_Workspace/THEIA**. Some older files contain stale claims. Prefer the current manuscript plus the later correction/review record for the question at hand. A source path is not a guarantee that every statement in that file remains current.

[P1] Current paper and long scientific record

revamp/campaign/results_v03_2026-09-23/manuscript/journal/MAIN_PAPER_v0.3.md

revamp/campaign/results_v03_2026-09-23/manuscript/MANUSCRIPT_DRAFT_v0.3.md

Use for the central question, model, historical methods and results. The journal paper is more current than the old paper/main.tex framing.

[P2] Claim-level boundaries

revamp/campaign/results_v03_2026-09-23/manuscript/journal/CLAIM_EVIDENCE_v0.3.md

Use to connect historical claims with estimands, source artifacts, denominators and permitted scope.

[P3] Generated historical N08 tables

revamp/campaign/parallel_expansion_2026-10-02/n08/readout/repro_discovery/N08_TABLES.md

Use for the five-cell results and paired contrasts. The plots in this guide transcribe these values; they are not new experimental runs.

[P4] Completed October 7 science crosswalk

revamp/campaign/execution_path_integration_2026-10-07_124017-2d0aea/science/CLAIM_CROSSWALK.md

Use for input-version scope, the 10-versus-9 sampling distinction and the last completed review's corrected-fit status.

[P5] Corrected replication contract

revamp/campaign/execution_path_integration_2026-10-07_124017-2d0aea/science/EXPERIMENT_CONTRACT.json

Use for the 50-fit matrix, primary interaction, seed handling and conditional secondary procedure. Later signed amendments may supersede this snapshot; verify before running or presenting.

33 | Source index and snapshot limits

[P6] Public-cohort qualification snapshot

revamp/campaign/execution_path_integration_2026-10-07_124017-2d0aea/data/ELIGIBILITY_FUNNEL.json

[P7] Evidence map and correction history

revamp/reviews/EVIDENCE_MAP.md

revamp/campaign/next_phase_2026-09-20/CLAIM_CORRECTIONS.md

[P8] Older study, historical context only

paper/main.tex and docs/MODEL_CARD.md

Do not use these files to overwrite the newer methods/claims. Some model-card descriptions and older plans are stale.

[P9] Codex review and subsequent direction

CODEX_REVIEW_EXECUTION_PATH_2026-10-07.md

The 10,000-patient requirement also comes directly from the user's project direction. It is not a scientific result.

Public sources checked on 7 October 2026

[W1] National Cancer Institute. Non-Small Cell Lung Cancer Treatment (PDQ), health professional version. Medical background only; no individual treatment advice.

<https://www.cancer.gov/types/lung/hp/non-small-cell-lung-treatment-pdq>

[W2] Society for Science. ISEF International Rules: Rules for All Projects, current 2027-cycle page.

<https://www.societyforscience.org/isef/international-rules/rules-for-all-projects/>

[W3] Society for Science. Grand Award Judging Criteria.

<https://www.societyforscience.org/isef/grand-award/criteria/>

[W4] JSHS. 2026-2027 Core Rules of Competition, especially eligibility and AI-use sections.

<https://www.jshs.org/wp-content/uploads/2026/08/JSHS-Core-Rules-of-Competition-2026-27.pdf>

[W5] JSHS. Oral Presentation Guidelines.

<https://www.jshs.org/national-symposium/oral-presentation-guidelines/>

This is a dated educational synthesis, not a fresh full scientific audit. Review corrected-input outcomes and official local competition requirements before preparing any final submission. Source hashes and document QA are stored alongside the editable guide source.